7

Plotting Points and Lines

Q. What did the dot say to the line?
A. Come to the point.

1. READING POINTS OFF A GRAPH

This chapter reviews some of the ideas about plotting points and lines which will be used in part III. You can either read this chapter now, or return to it if you run into difficulty in part III. If you read the chapter now, the first four sections are the most important; the last section is more difficult.

Figure 1 shows a horizontal axis (the x-axis) and a vertical axis (the y-axis). The point shown in the figure has an x-coordinate of 3, because it is in line with 3 on the x-axis. It has a y-coordinate of 2, because it is in line with 2 on the y-axis. This point is written $x = 3$, $y = 2$. Sometimes, it is abbreviated even more, to $(3, 2)$. The point shown in figure 2 is $(-2, -1)$: it is directly below -2 on the x-axis, and directly to the left of -1 on the y-axis.

Figure 1.

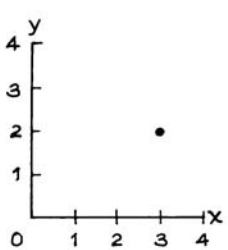
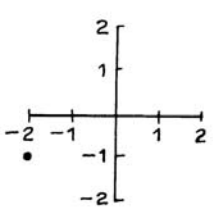


Figure 2.

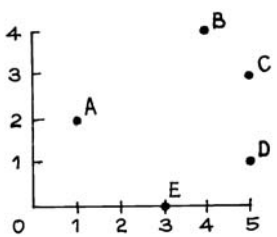


The idea of representing points by pairs of numbers is due to René Descartes (France, 1596–1650). In his honor, the x- and y-coordinates are often called "cartesian coordinates."

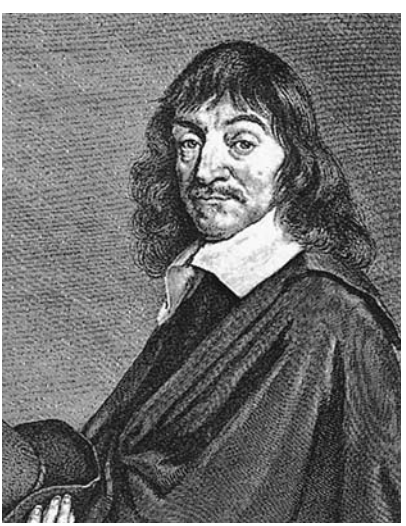
Exercise Set A

1. Figure 3 shows five points. Write down the x-coordinate and y-coordinate for each point.
2. As you move from point A to point B in figure 3, your x-coordinate goes up by _____; your y-coordinate goes up by _____.
3. One point in figure 3 has a y-coordinate 1 bigger than the y-coordinate of point E. Which point is that?

Figure 3.



The answers to these exercises are on p. A52.



René Descartes (France, 1596–1650)

Wolff-Leavenworth Collection, courtesy of the Syracuse University Art Collection.

2. PLOTTING POINTS

Figure 4 shows a pair of axes. To plot the point (2, 1), find the 2 on the x-axis. The point will be directly above this, as in figure 5. Find the 1 on the y-axis, the point will be directly to the right of this, as in figure 6.

Figure 4.

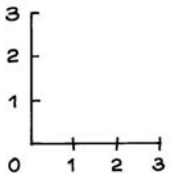


Figure 5.

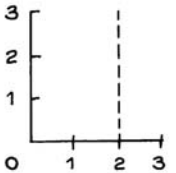
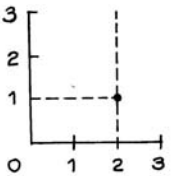


Figure 6.



Exercise Set B

1. Draw a pair of axes and plot each of the following points:

   (1, 1) (2, 2) (3, 3) (4, 4)

   What can you say about them?

2. Three out of the following four points lie on a line. Which is the maverick? Is it above or below the line?

   (0, 0) (0.5, 0.5) (1, 2) (2.5, 2.5)

3. The table below shows four points. In each case, the y-coordinate is computed from the x-coordinate by the rule $y = 2x + 1$. Fill in the blanks, then plot the four points. What can you say about them?

x	y
1	3
2	5
3	–
4	–

4. Figure 7 below shows a shaded region. Which of the following two points is in the region: (1, 2) or (2, 1)?

5. Do the same for figure 8.

6. Do the same for figure 9.

Figure 7.

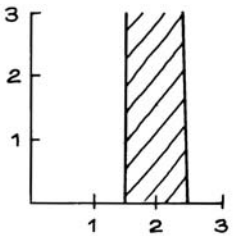


Figure 8.

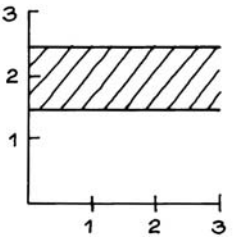
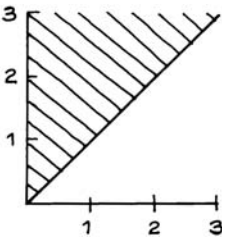


Figure 9.



3. SLOPE AND INTERCEPT

Figure 10 shows a line. Take any point on the line—for instance, point A. Now move up the line to any other point—for instance, point B. Your x-coordinate has increased by some amount, called the *run*. In this case, the run was 2. At the same time your y-coordinate has increased by some other amount, called the *rise*. In this case, the rise was 1. Notice that in this case, the rise was half the run. Whatever two points you take on this line, the rise will be half the run. The ratio rise/run is called the *slope* of the line:

$$\boxed{\text{slope} = \text{rise/run}.}$$

The slope is the rate at which y increases with x, along the line. To interpret it another way, imagine the line as a road going up a hill. The slope measures the steepness of the grade. For the line in figure 10, the grade is 1 in 2—quite steep for a road. In figure 11, the slope of the line is 0. In figure 12, the slope is -1. If the slope is positive, the line is going uphill. If the slope is 0, the line is horizontal. If the slope is negative, the line is going downhill.

Figure 10. Slope is 1/2.

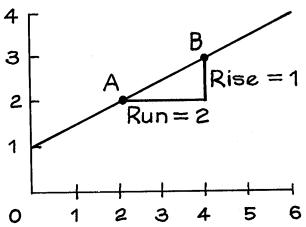


Figure 11. Slope is 0.

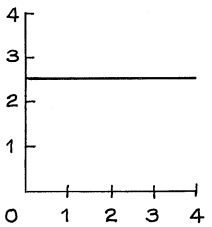
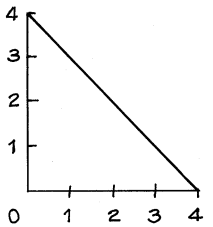


Figure 12. Slope is -1.



The *intercept* of a line is its height at $x = 0$. Usually, the axes cross at 0. Then, the intercept is where the line crosses the y-axis. In figure 13, the intercept is 2. Sometimes, the axes don't cross at 0, and then you have to be a little bit careful. In figure 14, the axes cross at (1, 1). The intercept of the line in figure 14 is 0—that would be its height at $x = 0$.

Often, the axes of a graph show units. For example, in figure 15 the units for the x-axis are inches, the units for the y-axis are degrees celsius. Then the slope and intercept have units too. In figure 15, the slope of the line is 2.5 degrees per inch; the intercept is -5 degrees.

Figure 13.

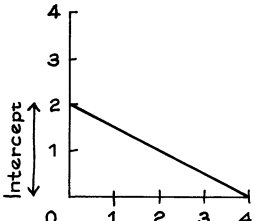


Figure 14.

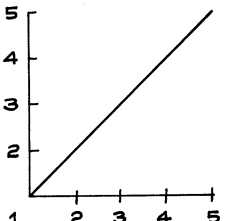
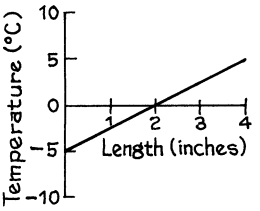


Figure 15.



Exercise Set C

1. Figures 16 to 18 show lines. For each line, find the slope and intercept. Note: the axes do not cross at 0 in each case.

Figure 16.

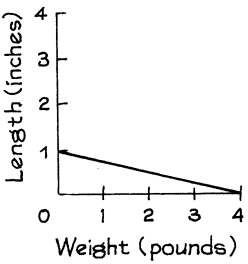


Figure 17.

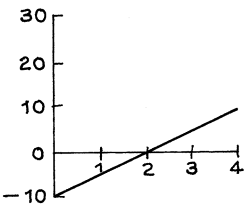
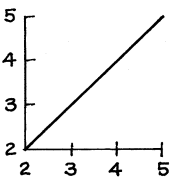


Figure 18.



The answers to these exercises are on p. A53.

4. PLOTTING LINES

Example 1. Plot the line which passes through the point (2, 1) and has slope 1/2.

Solution. First draw a pair of axes and plot the given point (2, 1), as in figure 19. Then move any convenient distance off directly to the right from the given point: figure 20 shows a run of 3. Make a construction point at this new location. Since the line slopes up, it passes above the construction point. How far? That is, how much will the line rise in a run of 3? The answer is given by the slope. The line is rising at the rate of half a vertical unit per horizontal unit, and in this case there is a run of 3 horizontal units, so the rise is $3 \times 1/2 = 1.5$:

$$\text{rise} = \text{run} \times \text{slope}.$$

Make a vertical move of 1.5 from the construction point, and mark a point at this third location, as in figure 21. This third point is on the line. Put a ruler down and join it to the given point (2, 1).

Figure 19.

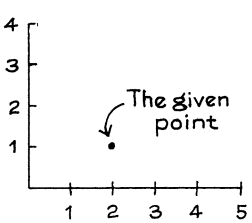


Figure 20.

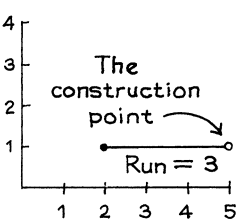
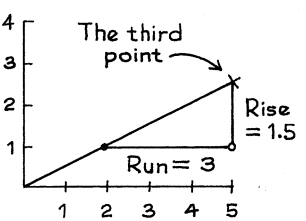


Figure 21.



Exercise Set D

1. Draw lines through the point (2, 1) with the following slopes:

   (a) $+1$ (b) -1 (c) 0

2. Start at the point (2, 1) in figure 21. If you move over 2 and up 1, will you be on the line, above the line, or below the line?

3. The same, but move over 4 and up 2.

4. The same but move over 6 and up 5.

5. Draw the line with intercept 2 and slope -1. Hint: this line goes through the point (0, 2).

6. Draw the line with intercept 2 and slope 1.

The answers to these exercises are on p. A54.

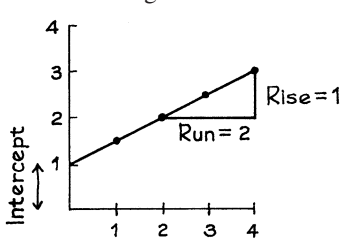
5. THE ALGEBRAIC EQUATION FOR A LINE

Example 2. Here is a rule for computing the y-coordinate of a point from its x-coordinate: $y = \frac{1}{2}x + 1$. The table below shows the points with x-coordinates of 1, 2, 3, 4. Plot the points. Do they fall on a line? If so, find the slope and intercept of this line.

Solution. The points are plotted in figure 22. They do fall on a line. Any point whose y-coordinate is related to its x-coordinate by the same equation $y = \frac{1}{2}x + 1$ will fall on the same line. This line is called the *graph* of the equation. The slope of the line is $\frac{1}{2}$, the coefficient of x in the equation. The intercept is 1, the constant term in the equation.

x	y
1	1.5
2	2.0
3	2.5
4	3.0

Figure 22.



> The graph of the equation $y = mx + b$ is a straight line, with slope m and intercept b.

Example 3. Figure 23 shows a line. What is the equation of this line? What is the height of this line at $x = 1$?

Solution. This line has slope -1 and intercept 4. Therefore, its equation is $y = -x + 4$. Substituting $x = 1$ gives $y = 3$; so the height of the line is 3 when x is 1.

Figure 23.

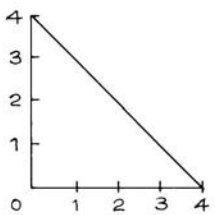


Figure 24.

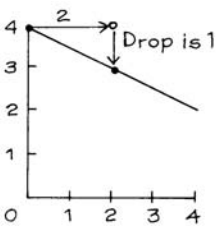
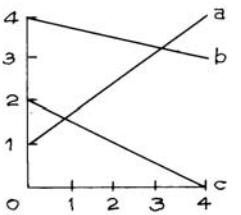


Figure 25.



Example 4. Plot the line whose equation is $y = -\frac{1}{2}x + 4$.

Solution. The intercept of this line is 4. Plot the point (0, 4) as in figure 24. The line must go through this point. Make any convenient horizontal move—say 2. The slope is $-1/2$, so the line must drop 1. Mark the point which is 2 over and 1 down from the first point in figure 24. Then join these two points by a straight line.

Exercise Set E

1. Plot the graphs of the following equations:

   (a) $y = 2x + 1$ (b) $y = \frac{1}{2}x + 2$

   In each case, say what the slope and intercept are, and give the height of the line at $x = 2$.

2. Figure 25 shows three lines. Match the lines with the equations:

$$y = \tfrac{3}{4}x + 1 \qquad y = -\tfrac{1}{4}x + 4 \qquad y = -\tfrac{1}{2}x + 2$$

3. Plot four different points whose y-coordinates are double their x-coordinates. Do these points lie on a line? If so, what is the equation of the line?

4. Plot the points (1, 1), (2, 2), (3, 3), and (4, 4) on the same graph. These points all lie on a line. What is the equation of this line?

5. For each of the following points, say whether it is on the line of exercise 4, or above, or below:

   (a) (0, 0) (b) (1.5, 2.5) (c) (2.5, 1.5)

6. True or false:

   (a) If y is bigger than x, then the point (x, y) is above the line of exercise 4.
   (b) If $y = x$, then the point (x, y) is on the line of exercise 4.
   (c) If y is smaller than x, then the point (x, y) is below the line of exercise 4.

The answers to these exercises are on pp. A54–55.